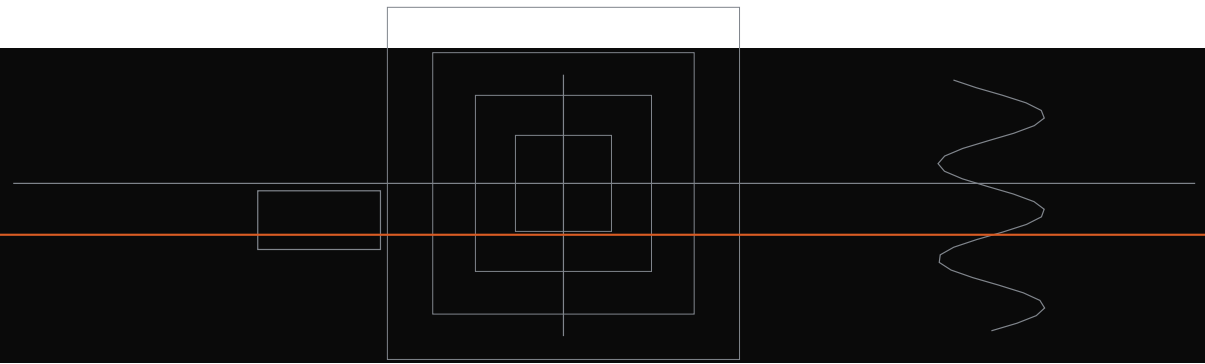


BILL OF MATERIALS

Robotic Arm

Rev. 2026-04-30 74 items



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Product/Project: Robotic Arm
Revision date: 2026-04-30

<https://www.printables.com/model/1510558-robotic-arm>

COMPONENT

33 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Nema 17 - Stepper Motor	1	AliExpress Link	Amazon Link	SM42HT47-1684 1,8" step, 1,6 A, 60mm
>>	Arduino Mega	1	AliExpress Link	Amazon Link	ARDUINO MEGA 2560
>>	Stepper Driver	x6	AliExpress Link	Amazon Link	Stepper Driver tb6560
>>	Power Supply	1	AliExpress Link	Amazon Link	Power Supply 24V 320W
>>	RAMPS Board	1	AliExpress Link	Amazon Link	RAMPS V1.4
>>	Voltage Converter	1	AliExpress Link	Amazon Link	Power converter 24V a 12V
>>	Servo Motor	1	AliExpress Link	Amazon Link	Servo Motor 180 deg 55G TORQUE=13KG/CM
>>	Nema 17 - Geared Stepper	1	AliExpress Link	Amazon Link	Gear Ratio 5:1 Planetary Gearbox (step 0,35" , 2Nm, 1,68A)
>>	Nema 23 - Stepper Motor	x2	AliExpress Link	Amazon Link	112mm Single shaft Ø8mm, step 1,8", 3A, 3Nm hoding torque
>>	Nema 14 - Stepper Motor	1	AliExpress Link	Amazon Link	SM35HT36-1004A, 1,8" step, 1A, 1400g*cm
>>	Nema 17 - Stepper Motor	1	AliExpress Link	Amazon Link	SM42HT33-1334, 1,8" step, 1,33A, 2200 g*cm
>>	Smooth Rod	1	AliExpress Link	Amazon Link	Smooth Rod_M8x134mm (cut to length)
>>	Smooth Rod	1	AliExpress Link	Amazon Link	Smooth Rod_M8x114mm (cut to length)
>>	Smooth Rod	1	AliExpress Link	Amazon Link	Smooth Rod_M8x80mm (cut to length)

COMPONENT continued

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Bearing	x10	AliExpress Link	Amazon Link	Ball Bearing 608ZZ 8mm x 22mm x 7mm
>>	Bearing	x8	AliExpress Link	Amazon Link	Ball Bearing 625ZZ 5mm x 16mm x 5mm
>>	Bearing	x9	AliExpress Link	Amazon Link	Ball Bearing 624ZZ 4mm x 13mm x 5mm
>>	Bearing	x3	AliExpress Link	Amazon Link	Ball Bearing 623ZZ 3mm x 10mm x 4mm
>>	Brass Heat Insert	x3	AliExpress Link	Amazon Link	M4x8x5
>>	Brass Heat Insert	x20	AliExpress Link	Amazon Link	M3x5,6x4,7
>>	Pulley	x3	AliExpress Link	Amazon Link	Pulley T5, Bore 8mm, 14 tooth, 17mm width
>>	Pulley	x2	AliExpress Link	Amazon Link	Pulley T5, Bore 5mm, 10 tooth, 17 mm width
>>	Threaded Rod	1	AliExpress Link	Amazon Link	ROD BAR M8 L 42mm (Order 50mm and grind 2mm off)
>>	Shaft Coupling	1	AliExpress Link	Amazon Link	COUPLING STEEL 5 TO 8 mm rigid
>>	Cooling Fan	1	AliExpress Link	Amazon Link	Axial Fan DC 24V 80x80x25mm
>>	Cooling Fan	1	AliExpress Link	Amazon Link	Axial Fan DC 24V 50x50x15mm
>>	Power Cable	1	AliExpress Link	Amazon Link	Power supply cable IEC 1,5m (buy the one for your region.
>>	USB Cable	1	AliExpress Link	Amazon Link	Cable USB 2.0 AM/BM 1,8m
>>	Cable Management	x10	AliExpress Link	Amazon Link	Cable Ties

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COMPONENT continued					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Bushing	x8	AliExpress Link	Amazon Link	Self Lubricating bearing 5mm x 8mm x 10mm
>>	Bushing	x2	AliExpress Link	Amazon Link	Self Lubricating bearing 8mm x 12mm x 20mm
>>	Timing Belt	1	AliExpress Link	Amazon Link	Breco belt T5, 15mm width, L 180cm
>>	Base Plate	1	-	-	Wood Base - 550mm x 550mm x 16mm (Or another material fo your choosing)

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FASTENERS

27 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Hexagonal Head Cap Screw	x4	AliExpress Link	Amazon Link	MC6S M3x8
>>	Hexagonal Head Cap Screw	x30	AliExpress Link	Amazon Link	MC6S M3x10
>>	Hexagonal Head Cap Screw	x6	AliExpress Link	Amazon Link	MC6S M3x12
>>	Hexagonal Head Cap Screw	x13	AliExpress Link	Amazon Link	MC6S M3x16
>>	Hexagonal Head Cap Screw	x4	AliExpress Link	Amazon Link	MC6S M3x20
>>	Hexagonal Head Cap Screw	x5	AliExpress Link	Amazon Link	MC6S M3x25
>>	Hexagonal Head Cap Screw	x2	AliExpress Link	Amazon Link	MC6S M3x30
>>	Hexagonal Head Cap Screw	x4	AliExpress Link	Amazon Link	MC6S M3x35
>>	Hexagonal Head Cap Screw	x7	AliExpress Link	Amazon Link	MC6S M3x40
>>	Hexagonal Head Cap Screw	x2	AliExpress Link	Amazon Link	MC6S M4x10
>>	Hexagonal Head Cap Screw	x2	AliExpress Link	Amazon Link	MC6S M4x12
>>	Hexagonal Head Cap Screw	x8	AliExpress Link	Amazon Link	MC6S M4x16
>>	Hexagonal Head Cap Screw	x3	AliExpress Link	Amazon Link	MC6S M4x20
>>	Hexagonal Head Cap Screw	x4	AliExpress Link	Amazon Link	MC6S M4x25
>>	Hexagonal Head Cap Screw	x8	AliExpress Link	Amazon Link	MC6S M4x30
>>	Hexagonal Head Cap Screw	x4	AliExpress Link	Amazon Link	MC6S M4x40
>>	Hexagonal Head Cap Screw	x6	AliExpress Link	Amazon Link	MC6S M4x45
>>	Hexagonal Head Cap Screw	x4	AliExpress Link	Amazon Link	MC6S M4x60
>>	Hexagonal Head Cap Screw	x120	AliExpress Link	Amazon Link	MC6S M5x14

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FASTENERS continued

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Hexagonal Head Cap Screw	x8	AliExpress Link	Amazon Link	MC6S M5x20
>>	Hexagonal Bolt	1	AliExpress Link	Amazon Link	M6S M8x65
>>	Nut	x30	AliExpress Link	Amazon Link	NUT M3
>>	Nut	x4	AliExpress Link	Amazon Link	NUT M4
>>	Locknut	x18	AliExpress Link	Amazon Link	LOCKNUT M4
>>	Locknut	x16	AliExpress Link	Amazon Link	LOCKNUT M5
>>	Locknut	x2	AliExpress Link	Amazon Link	LOCKNUT M8
>>	Washer	x10	AliExpress Link	Amazon Link	WASHER M3

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FILAMENTS 4 items					
	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	ABS/ASA/PETG&PC Filament	1	AliExpress Link	Amazon Link	Approx 500g required
>>	ABS/ASA/PETG&PC Filament	1	AliExpress Link	Amazon Link	Approx 500g required
>>	Black PLA Filament	1	AliExpress Link	Amazon Link	Approx 750g required (includes 500g for the stand)
>>	White TPU Filament	1	AliExpress Link	Amazon Link	Approx 50g required, for feet

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TOOLS

10 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Crimp Tool	1	AliExpress Link	Amazon Link	-
>>	Hex Key set	1	AliExpress Link	Amazon Link	-
>>	Soldering Iron	1	AliExpress Link	Amazon Link	-
>>	Wire cutter	1	AliExpress Link	Amazon Link	-
>>	Wire stripper	1	AliExpress Link	Amazon Link	-
>>	Precision Screwdriver Set	1	AliExpress Link	Amazon Link	-
>>	Heat insert set	1	AliExpress Link	Amazon Link	-
>>	Heat insert tip - set	1	AliExpress Link	Amazon Link	-
>>	Soldering Iron TS101	1	AliExpress Link	Amazon Link	-
>>	Heat Insert Kit	1	AliExpress Link	Amazon Link	-